BUSINESS TEMPLATE

Mountaineering Club Database

Logical Relational Model · PostgreSQL · 3NF

Legal Notice: This document contains privileged and/or confidential information and may not be disclosed, distributed or reproduced without the prior written permission of EPAM®.

Confidential

CONTENTS

BUSINESS DESCRIPTION 3
1.1 the Benefits of implementing a database. Project Vision 3
2 MODEL DESCRIPTION 3
2.1 Definitions & Acronyms 3
2.2 Logical Scheme 4
2.3 Objects 5

BUSINESS DESCRIPTION

1.1 THE BENEFITS OF IMPLEMENTING A DATABASE. PROJECT VISION

The Mountaineering Club needs a structured information system to manage its members, emergency contacts, mountains, expeditions, registrations, qualifications, experience records, equipment and safety incidents. When this information is maintained in separate spreadsheets or informal records, the club may face duplicated member data, inconsistent expedition registrations, incomplete safety information, difficulty tracking shared equipment and loss of historical status changes.

The proposed relational database provides a centralized and consistent source of information. It allows the club to register members once and reference them in expeditions, certifications, experience records and emergency-contact records. It also separates reusable reference data, such as countries, expedition types and certifications, from operational records.

Data integrity is supported through primary keys, foreign keys, NOT NULL, UNIQUE and CHECK constraints. Examples include unique member email addresses, unique country ISO codes, positive mountain elevations and expedition capacities, and an end date that cannot precede the start date. Foreign keys prevent registrations or incident records from referencing nonexistent members or expeditions.

Historical traceability is supported by EXPEDITION_STATUS_HISTORY, which stores each status period using valid_from and valid_to rather than overwriting the previous status. Equipment assignments also retain assignment and return timestamps and the condition of the equipment at issue and return. This makes it possible to review prior expedition and equipment events.

The model can scale by adding members, expeditions, mountains, equipment items, certifications and participation records without changing the core structure. New attributes can be added to the relevant entity, while new many-to-many relationships can be represented through associative tables. Constraints and indexes should be expanded as operational volume and query patterns grow.

2 MODEL DESCRIPTION

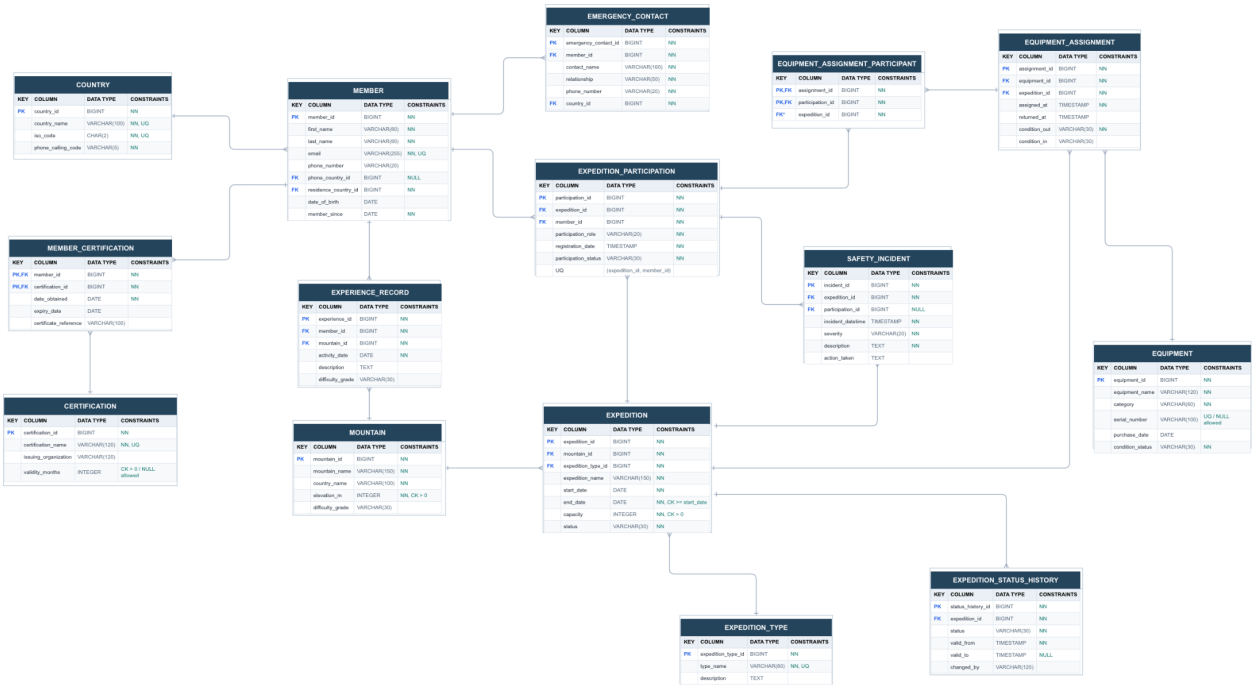
2.1 DEFINITIONS & ACRONYMS

TERM	DEFINITION
Member	A registered person who belongs to the club and may join expeditions.
Emergency contact	A person designated by a member to be contacted in an emergency.
Mountain	A mountaineering destination recorded by name, country, elevation and difficulty.
Expedition	A planned club activity taking place at a mountain during a defined date range.
Expedition participation	The association between a member and an expedition, including role, registration date and participation status.
Certification	A qualification issued by an organization, such as a climbing or first-aid credential.
Member certification	A record that a particular member holds a particular certification, including obtained and expiry dates.
Experience record	A record of a member's previous activity or

	experience at a mountain.
Equipment assignment	The allocation of one equipment item to one expedition for a time period.
Equipment assignment participant	Associates an equipment assignment with one or more participants, allowing equipment to be shared.
Safety incident	An event recorded during an expedition, optionally associated with a specific participant.
Status history	A time-bounded record of an expedition's status changes.
PK	Primary Key: uniquely identifies each row.
FK	Foreign Key: references a key in another table.
UQ	Unique constraint: prevents duplicate values or combinations.
NN	Not Null: requires a value.
CK	Check constraint: validates a condition on stored values.
NULL	The value is optional or currently unknown; it is not the same as an empty string.
3NF	Third Normal Form: non-key attributes depend on the key, the whole key and nothing but the key.

2.2 LOGICAL SCHEME

The following diagram represents the proposed logical relational schema. It contains tables and includes associative entities for the many-to-many relationships between members and expeditions, members and certifications, and shared equipment assignments and expedition participants.



The most challenging modeling decision was representing shared equipment. An equipment item is assigned to an expedition, not directly to one person. The bridge table EQUIPMENT_ASSIGNMENT_PARTICIPANT then associates that assignment with one or more expedition participation records. This avoids duplicating the same equipment assignment when several members

share an item.

The bridge table includes expedition_id to support a composite referential rule: the equipment assignment and the participant's participation must belong to the same expedition. To enforce this in PostgreSQL, the referenced tables must expose matching UNIQUE keys, such as (assignment_id, expedition_id) in EQUIPMENT_ASSIGNMENT and (participation_id, expedition_id) in EXPEDITION_PARTICIPATION, and the bridge table must define the corresponding composite foreign keys.

The model prevents common anomalies. A duplicate email is rejected by UNIQUE on MEMBER.email. A participation cannot refer to a nonexistent member or expedition because of its foreign keys. The unique combination (expedition_id, member_id) prevents registering the same member twice for one expedition. Separate MEMBER_CERTIFICATION rows prevent repeating member details for every qualification. Status history prevents the previous expedition status from being lost when a new status is recorded.

The structure remains scalable because new participants are inserted as rows rather than new columns, and new certification types or expedition types can be inserted as reference records. If requirements expand, the areas most likely to change are equipment allocation rules, geographic reference data for mountains, and the detail captured for safety incidents. These changes can be introduced through additional attributes or related tables.

2.3 OBJECTS

The following descriptions summarize the purpose, normalization, key choices and principal attributes of each table in the submitted diagram.

COUNTRY

Stores country reference data used for member residence, phone calling codes and emergency-contact phone country.

Attributes and data types:

FIELD DEFINITION
country_id BIGINT PK
country_name VARCHAR(100) NN UQ
iso_code CHAR(2) NN UQ
phone_calling_code VARCHAR(5) NN.

Normalization level: 3NF. Each row represents one country, and all non-key attributes (name, ISO code and calling code) describe that country. Country name and ISO code are alternate unique identifiers; no non-key attribute determines another non-key attribute in this table.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
1	Colombia	CO	+57
2	Canada	CA	+1

MEMBER

Stores each club member's identity and contact information.

Attributes and data types:

FIELD DEFINITION
member_id BIGINT PK
first_name VARCHAR(80) NN
last_name VARCHAR(80) NN
email VARCHAR(255) NN UQ
phone_number VARCHAR(20)
phone_country_id BIGINT FK NULL
residence_country_id BIGINT FK NN
date_of_birth DATE
member_since DATE NN.

Normalization level: 3NF. Each row represents one member. Personal and contact attributes depend on member_id. Country details are not duplicated in this table; they are referenced through foreign keys to COUNTRY. The two country FKs have different meanings (residence and phone calling code).

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
101	Ana	Pérez	ana@example.org
102	Luis	Gómez	luis@example.org

EMERGENCY_CONTACT

Stores emergency contacts nominated by members.
Attributes and data types:

FIELD DEFINITION
emergency_contact_id BIGINT PK
member_id BIGINT FK NN
contact_name VARCHAR(160) NN
relationship VARCHAR(50) NN
phone_number VARCHAR(20) NN
country_id BIGINT FK NN.

Normalization level: 3NF. Each row represents one contact for one member. Contact details depend on emergency_contact_id; member_id references the owner. Multiple contacts are stored as separate rows rather than repeating contact columns in MEMBER.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
501	101	Marta Pérez	Mother
502	101	Juan Pérez	Brother

MOUNTAIN

Stores mountaineering destinations and their physical characteristics.

Attributes and data types:

FIELD DEFINITION
mountain_id BIGINT PK
mountain_name VARCHAR(150) NN
country_name VARCHAR(100) NN
elevation_m INTEGER NN CK > 0
difficulty_grade VARCHAR(30).

Normalization level: 3NF within the current design. Mountain attributes depend on mountain_id. country_name is stored as a descriptive value; if countries are controlled centrally, replace it with country_id referencing COUNTRY to strengthen referential integrity.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
10	Mount Example	Colombia	5100
11	Peak Example	Peru	6200

EXPEDITION_TYPE

Defines reusable categories of expedition.

Attributes and data types:

FIELD DEFINITION
expedition_type_id BIGINT PK
type_name VARCHAR(80) NN UQ
description TEXT.

Normalization level: 3NF. Each row represents one reusable expedition type. type_name and description depend on expedition_type_id; type_name is unique.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2
1	Trekking
2	Technical climbing

EXPEDITION

Stores each planned or completed expedition.

Attributes and data types:

FIELD DEFINITION
expedition_id BIGINT PK
mountain_id BIGINT FK NN
expedition_type_id BIGINT FK NN
expedition_name VARCHAR(150) NN
start_date DATE NN
end_date DATE NN CK >= start_date
capacity INTEGER NN CK > 0
status VARCHAR(30) NN.

Normalization level: 3NF. Each expedition's destination, type, dates, capacity and status depend on expedition_id. Mountain and expedition-type descriptions are kept in their own tables and referenced by FKs.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
1001	Andes weekend	10	1
1002	Summit challenge	11	2

EXPEDITION_PARTICIPATION

Associates members with expeditions and records each person's role and registration state.

Attributes and data types:

FIELD DEFINITION
participation_id BIGINT PK
expedition_id BIGINT FK NN
member_id BIGINT FK NN
participation_role VARCHAR(20) NN
registration_date TIMESTAMP NN
participation_status VARCHAR(30) NN
UNIQUE(expedition_id, member_id).

Normalization level: 3NF. This associative entity resolves the M:N relationship between MEMBER and EXPEDITION. Its attributes describe the participation. UNIQUE(expedition_id, member_id) prevents duplicate membership in the same expedition.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
9001	1001	101	PARTICIPANT
9002	1001	102	GUIDE

CERTIFICATION

Stores the catalog of qualifications that members may hold.

Attributes and data types:

FIELD DEFINITION
certification_id BIGINT PK
certification_name VARCHAR(120) NN UQ
issuing_organization VARCHAR(120)
validity_months INTEGER CK > 0 or NULL.

Normalization level: 3NF. Certification name, issuer and validity period describe the certification identified by certification_id. Member-specific dates are stored separately in MEMBER_CERTIFICATION.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2
1	First Aid
2	High-altitude Safety

MEMBER_CERTIFICATION

Records which certifications each member holds, including dates and certificate reference.

Attributes and data types:

FIELD DEFINITION
member_id BIGINT PK/FK
certification_id BIGINT PK/FK
date_obtained DATE NN
expiry_date DATE
certificate_reference VARCHAR(100).

Normalization level: 3NF. This associative table resolves the M:N relationship between MEMBER and CERTIFICATION. The composite key (member_id, certification_id) identifies a member's certification record; obtained and expiry dates depend on the complete key.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3
101	1	2025-03-10
102	2	2026-01-15

EXPERIENCE_RECORD

Stores a member's previous mountaineering experience at a mountain.

Attributes and data types:

FIELD DEFINITION
experience_id BIGINT PK
member_id BIGINT FK NN
mountain_id BIGINT FK NN
activity_date DATE NN
description TEXT
difficulty_grade VARCHAR(30).

Normalization level: 3NF. Each row is a distinct experience event. Its date, description and difficulty depend on experience_id, while member and mountain are referenced through FKs.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
7001	101	10	2025-08-12
7002	102	11	2026-02-04

EQUIPMENT

Stores the club's inventory of equipment items.

Attributes and data types:

FIELD DEFINITION
equipment_id BIGINT PK
equipment_name VARCHAR(120) NN
category VARCHAR(60) NN
serial_number VARCHAR(100) UQ or NULL
purchase_date DATE
condition_status VARCHAR(30) NN.

Normalization level: 3NF. Each row represents one inventory item. Name, category, serial number, purchase date and condition describe equipment_id. Assignment history is stored separately so it does not overwrite inventory facts.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3
301	Climbing rope	Rope
302	Helmet	Protection

EQUIPMENT_ASSIGNMENT

Records the assignment of one equipment item to one expedition, including issue and return condition.

Attributes and data types:

FIELD DEFINITION
assignment_id BIGINT PK
equipment_id BIGINT FK NN
expedition_id BIGINT FK NN
assigned_at TIMESTAMP NN
returned_at TIMESTAMP
condition_out VARCHAR(30) NN
condition_in VARCHAR(30).

Normalization level: 3NF. Each row represents one assignment event for one equipment item and one expedition. Assignment timestamps and condition details depend on assignment_id; equipment and expedition are referenced by FKs.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
8001	301	1001	2026-06-01
8002	302	1001	2026-06-01

EQUIPMENT_ASSIGNMENT_PARTICIPANT

Links one equipment assignment to one or more expedition participants, representing shared use.

Attributes and data types:

FIELD DEFINITION
assignment_id BIGINT PK/FK
participation_id BIGINT PK/FK
expedition_id BIGINT FK* NN.

Normalization level: 3NF. This bridge table resolves the M:N association between equipment assignments and expedition participations. The composite key (assignment_id, participation_id) identifies each association. expedition_id supports composite FKs that enforce same-expedition consistency.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3
8001	9001	1001
8001	9002	1001

SAFETY_INCIDENT

Stores safety incidents occurring during an expedition and optionally identifies an involved participant.

Attributes and data types:

FIELD DEFINITION
incident_id BIGINT PK
expedition_id BIGINT FK NN
participation_id BIGINT FK NULL
incident_datetime TIMESTAMP NN
severity VARCHAR(20) NN

description TEXT NN
action_taken TEXT.

Normalization level: 3NF. Each row represents one incident. Date/time, severity, description and action depend on incident_id. participation_id is nullable because an incident may not be attributed to a particular participant.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
6001	1001	9001	Minor
6002	1001	NULL	Moderate

EXPEDITION_STATUS_HISTORY

Stores the sequence of status periods for an expedition.

Attributes and data types:

FIELD DEFINITION
status_history_id BIGINT PK
expedition_id BIGINT FK NN
status VARCHAR(30) NN
valid_from TIMESTAMP NN
valid_to TIMESTAMP NULL
changed_by VARCHAR(120).

Normalization level: 3NF. Each row represents one status period or historical change for an expedition. Status and validity timestamps depend on status_history_id, preserving past values instead of overwriting them.

Illustrative example records (abbreviated):

VALUE 1	VALUE 2	VALUE 3	VALUE 4
1	1001	PLANNED	2026-05-01
2	1001	COMPLETED	2026-06-03